



External Semantics for Argumentation Frameworks

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External semantics

We propose external semantics to require defence from outside the attacked argument itself.

When an accepted argument is attacked, another argument in the argumentation set should answer the attacker. This blocks self defence while preserving third-party defence in even-cycle structures.

1. Motivation

- In practice, self defence is often not persuasive enough.
- In symmetric argumentation frameworks, Dung-admissible sets collapse into conflict-free sets, while strong admissibility can initially accept only isolated arguments.

2. External Defence and Semantics

- We require **third-party defence**.
- External semantics are built from **external defence**.

3. Topological Comparison

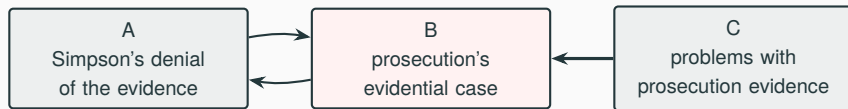
- Dung admissibility, external admissibility, and strong admissibility differ in how they treat **even cycles**.

Motivation

Courtroom motivation

O.J. Simpson trial

- Simpson directly denied the prosecution's evidential case.
- The defence also raised problems in the prosecution's evidence.
- These evidence problems gave **third-party support** to Simpson's denial.

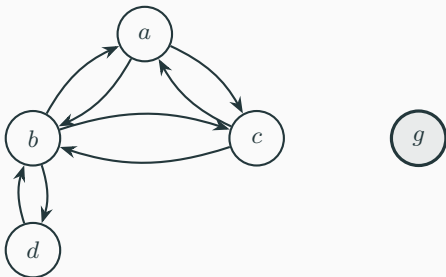


Formal reading

$$B \multimap A, \quad A \multimap B, \quad C \multimap B, \quad C \neq A.$$

A 's answer to B is still self defence. The external support comes from C , a distinct argument exposing problems in the evidence offered by the prosecution.

Dilemma in symmetric argumentation frameworks



Dung admissibility

Every attack gives the target a counterattack.
Thus every conflict-free set becomes self
defending:

$$ADF(SAF) = CF(SAF).$$

Strong admissibility

Defence must start from unattacked
arguments. In symmetric argumentation
frameworks, every non-isolated argument
loses that root.

Dung admissibility is too broad, while strong admissibility is too narrow.

Existing semantics also cannot solve the self defence problem in this setting.

External defence and semantics

For an argumentation framework $AF = \langle Arg, \rightarrow \rangle$ and an argumentation set $\Delta \subseteq Arg$:

Definition

$$\Delta \text{ } \epsilon\text{-defends } C \text{ iff } \forall B, B \rightarrow C \Rightarrow \Delta \setminus \{C\} \rightarrow B.$$

External defence function

$$d_{\epsilon}(\Delta) = \{C \in Arg \mid \Delta \text{ } \epsilon\text{-defends } C\}.$$

External admissibility

Δ is ϵ -admissible iff it is conflict-free and

$$\Delta \subseteq d_{\epsilon}(\Delta).$$

Dung, External, and Strong Admissibility

Admissibility	Intuition	Requirement
Dung admissibility	Allows an accepted argument to defend itself.	Conflict-free, and every member is defended by Δ .
External admissibility	Requires third-party defence from another accepted argument.	Conflict-free, and every member is defended by $\Delta \setminus \{A\}$.
Strong admissibility	Requires a defence chain traceable to unattacked arguments.	Each accepted argument has a finite defence chain rooted in unattacked arguments.

External admissibility blocks self defence without demanding an unattacked root.

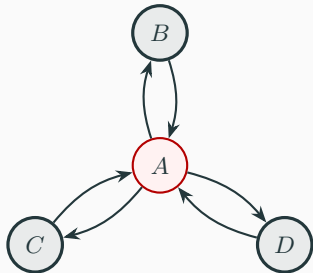
External admissibility's position on the admissibility spectrum

$$sADF(AF) \subsetneq \epsilon\text{-}ADF(AF) \subsetneq ADF(AF)$$

- Dung admissibility accepts an argument that defends itself.
- External admissibility requires third-party defence and rejects self defence.
- Strong admissibility requires a defence chain rooted in an unattacked argument.

A four-argument even-cycle example

The star pattern isolates the even-cycle issue



Admissibility	Admissible sets
Dung	$\emptyset$, $\{A\}$, and non-empty $\Gamma \subseteq \{B, C, D\}$
External	$\emptyset$, plus all $\Gamma \subseteq \{B, C, D\}$ with $ \Gamma \geq 2$
Strong	$\emptyset$ only

Why leaf sets work externally

Any two leaves among B, C, D can defend each other against A . The set $\{B, C, D\}$ works for the same reason.

Extensions built from d_ϵ

- Δ is **ϵ -complete** iff Δ is conflict-free and $\Delta = d_\epsilon(\Delta)$.
- Δ is **ϵ -preferred** iff Δ is a maximal ϵ -admissible set.
- Δ is **ϵ -grounded** iff Δ is the least fixed point of d_ϵ .

Basic results

- every finite framework has an ϵ -preferred extension.
- the ϵ -grounded extension coincides with Dung's grounded extension.
- stable extensions are unchanged, but they no longer guarantee ϵ -preferredness.

Topological Comparison

Even-cycle pressure

Odd cycles are not the main contrast

In an odd cycle with no attacks from outside the cycle, Dung admissibility, external admissibility, and strong admissibility reject every argument in that cycle. Many other semantics vary on odd cycles.

Symmetric argumentation frameworks as a representative case

A symmetric argumentation framework satisfies $A \rightarrow B$ iff $B \rightarrow A$. They are a special case where mutual attacks make even-cycle defence visible.

The even-cycle question

Admissibility	Treatment of even-cycle defence
Dung	in an unattacked 2-cycle, each argument can defend itself and be accepted on its own
External	rejects that unattacked 2-cycle self defence. In a larger unattacked even cycle, arguments in the cycle can defend one another as third-party defenders
Strong	rejects every argument in an unattacked even cycle

- It occupies a strict middle position:

$$sADF(AF) \subsetneq \epsilon\text{-}ADF(AF) \subsetneq ADF(AF).$$

- Dung admissibility, external admissibility, and strong admissibility treat odd cycles in the same way.
- Their different attitudes toward **even cycles** distinguish the three semantics.

Open direction

- Does third-party corroboration fully capture the epistemic ideal?
- Weighted or probabilistic defence raises a broader question about expressive power.

Thank you!

Questions and comments are welcome.